DURGA PRASAD SAHOO

Full-Stack AI Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience in Machine Learning, Deep Learning, Generative AI, and LLM applications using Python, RAG, LangChain, LangGraph, and OpenAI. Skilled in building end-to-end AI solutions, REST APIs, and full-stack applications with React, Node.js, and Flask, with experience achieving 0.96 ROC AUC in ML classification.

EDUCATION

B.Tech in Computer Science & Engineering — GIET University, Rayagada, Odisha 8.4/10 *Jul 2026*

Higher Secondary (12th) — Hi-Tech Higher Secondary School, Jajpur, Odisha 80.80%*Jul 2022*

PROFESSIONAL EXPERIENCE

Full Stack Developer & AI Engineer Intern — Sonydig Technology Pvt. Ltd. *Dec 2025 – Mar 2026*

Bhubaneswar, India | Odisha State Archaeology — Government Project under Dept. of OLL&C

- Implemented structured data handling, input validation, and optimized MySQL database queries, improving application performance, reliability, and user interaction.
- Researched and developed a SaaS-based AI-powered E-Office platform integrating an AI chatbot, OCR-enabled document processing, and an image generation model.
- Researched and fine-tuned a custom-trained Stable Diffusion image generation model to extend platform functionality and enhance user experience.

MERN Developer Intern — Vanur-Tech Media Pvt. Ltd. *May 2025 – Jun 2025*

Bhubaneswar, India

- Rebuilt the LMS platform UI by architecting reusable React components, improving cross-device accessibility and reducing redundant code across course modules.
- Partnered with a 3-person team to audit and optimize UI performance across the LMS platform, ensuring consistent rendering and responsiveness on mobile, tablet, and desktop.

PROJECTS

Auto Insurance Fraud Claim Detection | Python, Scikit-learn, Random Forest, Pandas, NumPy, Streamlit *2026*

- Built an end-to-end machine learning pipeline to detect fraudulent auto insurance claims, training a Random Forest Classifier with balanced class weighting on imbalanced claim data, achieving 88.42% accuracy and 0.96 ROC AUC.
- Designed a preprocessing pipeline (feature selection, date handling, one-hot encoding, feature-column alignment) to transform raw policy, accident, and vehicle data into model-ready features.
- Deployed a real-time fraud risk-scoring Streamlit web application and an analytics dashboard visualizing fraud trends, geographic patterns, and risk factors for claim investigators.

Customer Refund Management using Agentic AI | React.js, MongoDB, RAG, OpenAI, LangChain, LangGraph *Mar 2026*

- Built a full-stack AI customer support platform automating refund eligibility checks and customer conversations using Retrieval-Augmented Generation (RAG) and LLM agents.
- Developed text and voice AI workflows using Groq Whisper STT and ElevenLabs TTS, with real-time admin monitoring via WebSocket-based agent reasoning logs.

Nova Stay — AI-Powered Reservation System | Python, Flask, RAG, GenAI, ChromaDB, Hugging Face *Sep 2025*

- Deployed a Flask backend integrating OpenAI and Hugging Face LLMs to power a context-aware chatbot for hotel operations, achieving over 90% intent accuracy.
- Automated booking confirmation, GST-compliant invoice generation, and stay-duration calculation, eliminating manual invoice errors.

- Designed modular prompt-engineering workflows for scalable, maintainable API performance across 50+ hotel operation scenarios.

Uber Application | React.js, Tailwind CSS, Express.js, Node.js, MongoDB, Socket.IO, JWT *May 2025*

- Implemented JWT-based authentication and authorization to securely manage rider and driver sessions.
- Applied Dijkstra's algorithm to compute optimal routes, improving fuel efficiency by up to 20%.
- Built real-time bidirectional communication between riders and drivers using Socket.IO.

TECHNICAL SKILLS

Programming Languages : Python, JavaScript (ES6+), C, SQL

Full-Stack Development : Node.js, Express.js, React.js, REST APIs, JWT, RBAC, WebSocket, MongoDB, MySQL, PostgreSQL

Machine Learning & AI : Machine Learning, Deep Learning, Natural Language Processing (NLP), Random Forest, Classification Models, Model Training, Evaluation & Tuning, Feature Engineering

Generative AI & LLM : Generative AI, Agentic AI, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Prompt Engineering, OpenAI API, Hugging Face, LLM Integration

Frameworks & Libraries : Scikit-learn, TensorFlow, Pandas, NumPy, Flask, Streamlit

Vector Databases : Pinecone, ChromaDB. FAISS

Tools & Platforms : Git, GitHub, Postman, Anaconda, Jupyter Notebook

Core CS : Data Structures & Algorithms, OOPs, System Design

CERTIFICATIONS

- 4-Star Certification in Python — HackerRank
- Level One Swag, Google Cloud Challenges — Cloud Knowledge Recognition
- Wipro Java FullStack Course